# Optimal Symbol Rate for Discrete Nonlinear Frequency Division Multiplexing Transmissions

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**Abstract** *By analyzing the noise power dependence on symbol rate in the discrete eigenvalue transmission, the optimal symbol rate the minimizes the noise power is analytically found to be low, and the possible gain of advanced equalization algorithm is characterized.* *©2025 The Author(s)*

## Introduction

Nonlinear Frequency Division Multiplexing (NFDM) [1] is a promising theory that regards the fiber nonlinearity as an inherent system property rather than a destructive factor, because if the fiber is lossless, by which the nonlinear Schrödinger equation (NLSE) is transformed into linear evolution of the nonlinear spectrum, while the eigenvalues are kept invariant.

As real fiber is always lossy, the integrability of NLSE is broken. Besides, to compensate for the loss, optical amplifiers are introduced, which also introduce noise during transmission. Consequently, the eigenvalue and the nonlinear spectrum will become noisy, and the signal cannot be recovered by simply back rotating the discrete nonlinear spectrum by the full transmission distance since the eigenvalues are contaminated with noise.

Many efforts have been made to narrow the gap between the "lossless" assumption and the reality, including using Raman amplification [2]to reduce the power fluctuations along the fiber link, using linear minimum mean square error (LMMSE) estimator [3] for denoising on the nonlinear domain, nonlinear filters and machine learning [4] are also applied in this field. Recently, it has been proposed that back rotating the noisy eigenvalue by half of the transmission distance optimizes the SNR of b-coefficient [5, 6].

In this paper, we investigated the fundamental limitation of noise power of b-coefficient and eigenvalue on the overall noise. We numerically study the dependence of overall noise on symbol rate and determine an optimal symbol rate for discrete eigenvalue transmission system.

## NFT and noise evolution model

The analysis and result apply to a dual polarization discrete system, however, we discuss in a single polarization condition for simplicity, where the signal propagation is described by the normalized NLSE, $jq_z + q_{tt}/2 + |q|^2 q = 0$. The nonlinear spectrum of $q(0,t)$ is obtained by solving the eigenvalue problem, and the nonlinear Fourier coefficients $a(\lambda)$ and $b(\lambda)$ are given by[1]

$$\frac{\partial \boldsymbol{v}}{\partial t} = \begin{pmatrix} -j\lambda & jq(0,t) \\ jq^*(0,t) & j\lambda \end{pmatrix} \boldsymbol{v} \tag{1}$$

$$\begin{bmatrix} a(\lambda) \\ b(\lambda) \end{bmatrix} = \lim_{t\to+\infty} \begin{bmatrix} v_1 e^{j\lambda t} \\ v_2 e^{-j\lambda t} \end{bmatrix} \text{ with } \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \xrightarrow{t\to-\infty} \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-j\lambda t}$$

The discrete spectrum is defined as $Q_d(\lambda_i) = b(\lambda_i)/a'(\lambda_i)$ where $\lambda_i \in \mathbb{C}^+$ is the solutions of $a(\lambda_i) = 0$. In this paper, we restrict our discussion in a single discrete eigenvalue system, $i = 1$. Information is encoded on the discrete b-coefficient $b(\lambda_i)$, whose evolution is governed by $b(L,\lambda_i) = b(0,\lambda_i)\exp(-2jL\lambda_i^2)$. (We will use $i$ to denote span number hereafter, rather than the ordinal of eigenvalue as there is only one eigenvalue)

In an EDFA transmission system shown in Fig. 1, due to the perturbation of amplified spontaneous emission (ASE) noise from each EDFA, the evolution of $\lambda_i$ may be modelled as a discrete random walk, and the evolution of $b_i$ is an accumulation of the random walk of $\lambda_i$ times a productive noise $e^{\epsilon_i}$, given by $b_{i+1} = b_i e^{-2jL\lambda_i^2} e^{\epsilon_i}$, where $\epsilon_i \in \mathbb{C}$ [8].

It would be better to work with the logarithm of $b$, by which the productive noise is transferred into additive noise and hence the noise variance dependence on the magnitude of signal can be removed. Let $B_i = \ln(b_i)$ so that

$$B_{i+1} = B_i + \underbrace{(-2jL\lambda_i^2)}_{\Delta N_i^t} + \underbrace{\epsilon_i}_{\Delta N_i^d} \tag{2}$$

$$B_{i+1} = B_0 + \underbrace{\left(-2jL\sum_{i=1}^{N}\lambda_i^2\right)}_{N_{i+1}^t} + \underbrace{\sum_{i=1}^{N}\epsilon_i}_{N_{i+1}^d} \tag{3}$$

where $\lambda_i = \alpha_i + j\beta_i$ and $b_i$ denote the eigenvalue and b-coefficient right after the *i*-th EDFA. $L = L_A/L_D$ is the normalized span length with respect to the dispersion length. By taking logarithm, an APSK constellation commonly used in discrete NFDM transmission is transformed into a QAM constellation, as illustrated in Fig. 2.

We classify the noise terms as "Transfer Noise" ($\Delta N_i^t$) and "Direct Noise" ($\Delta N_i^d$), as indicated in Eq. 2, and the accumulated transfer noise and direct noise up to the *i*-th EDFA are denoted by $N_i^t$ and $N_i^d$, respectively. It reduces to

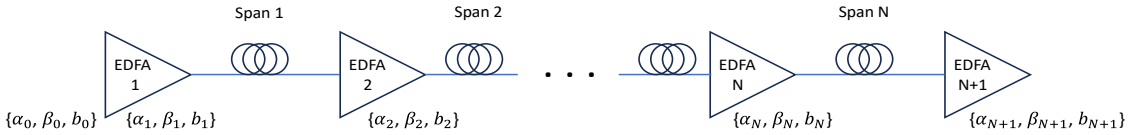


**Fig. 1:** Schematic diagram of the transmission link. ASE noise is independently added to λ and *b* right after each EDFA.

the model in [6] if the direct noise is ignored. To investigate the possible fundamental noise limitation on the performance of discrete eigenvalue transmission system, we carried out ideal back rotation assuming $\lambda_i$ is known for all $i$, given by $B_k^{BRI} = B_{N+1} - 2jL\sum_{i=k}^{N}\lambda_i^2$. After a full ideal back rotation, transfer noise is fully removed, and only the accumulated direct noise remains. We also conduct practical back rotation given by $B_k^{BRP} = B_{N+1} - 2jL(N+1-k)\lambda_{N+1}^2$ that assumes the receiver only has access to the received eigenvalue.

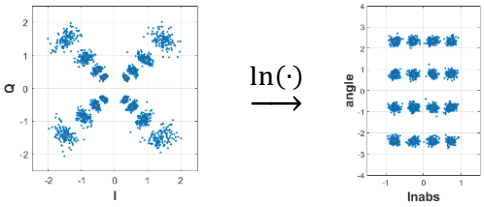


**Fig. 2:** APSK transformed into QAM constellation.

## Impact of symbol rate on the overall noise

Given a fixed span length and duty cycle $D_c$ (soliton FWHM/symbol period), we may vary the span number ($N$) and symbol rate ($R_S$) to investigate the dependence of $B$ noise power on them.

In the time domain, SNR is given by

$$SNR = \frac{P_S}{P_N} \propto \frac{E_S R_S}{BW N_0} \propto \frac{E_S}{N_0} \propto \frac{R_S}{N_0} \qquad (4)$$

where $P_S$, $P_N$ denote the power of soliton signal and in-band ASE noise, and $E_S$, $BW$ denote the energy, bandwidth of the soliton pulse, respectively, and we used the fact that the energy of a soliton is inversely proportional to its pulse duration. $N_0$ represents the power spectral density of ASE noise.

We assume that under the small noise perturbation condition, the time domain SNR linearly translated into NFT domain. In other words, the power of noise added by each EDFA is proportional to $R_s^{-1}$. Let

$$\mathrm{Var}(\Delta\lambda_i) = p_1 R_s^{-1}, \qquad \mathrm{Var}(\Delta N_i^d) = p_2 R_s^{-1} \qquad (5)$$

where $p_1$ and $p_2$ are scaling constants. Physically, $\Delta\lambda_i$ and $\Delta N_i^d$ means the amplitude, frequency phase and time position of a soliton pulse are all slightly shifted due to ASE loading of each EDFA. By increasing symbol rate, a higher SNR is obtained, and hence less amount of $\Delta\lambda_i$ and $\Delta N_i^d$. However, the dispersion length $L_D \propto R_s^{-2}$, and such a dependence will in turn dramatically amplify the $R_s^{-1}$-dependent noise on $\lambda_i$ and make the transfer noise $R_s^3$-dependent, which dominates the overall noise power and severely degrade the performance.

The noise variance of $B_i$ is $\mathrm{Var}(B_i) = \mathrm{Var}(N_i^t + N_i^d) = \mathrm{Var}(N_i^t) + \mathrm{Var}(N_i^d)$, where we have assumed the transfer noise and direct noise is uncorrelated (which can be verified by simulations). Considering the perturbated eigenvalue is still around $0 + 0.5j$, i.e., $\alpha_i \approx 0$, $\beta_i \approx 0.5$, then $\lambda_i^2 \approx 0.25 - \beta_i + j\alpha_i$, thus $\mathrm{Var}(\lambda_i^2) \approx \mathrm{Var}(\lambda_i)$. According to Eq. 9 and 10 in [6], we obtain,

$$\begin{aligned}\mathrm{Var}(N_{N+1}^t) &= 4L^2N^3/3 \cdot \mathrm{Var}(\Delta\lambda_i) \\ &= 4(L_A/L_D)^2N^3/3 \cdot p_1 R_s^{-1} \\ &= p_1 c N^3 R_s^3 \end{aligned} \qquad (6)$$

$$\mathrm{Var}(N_{N+1}^d) = N\mathrm{Var}(\Delta N_i^d) = p_2 N R_s^{-1} \qquad (7)$$

where $c = (4/3)1.736^4 L_A^2\beta_2^2/D_c^4$, a constant independent on symbol rate. Therefore,

$$\mathrm{Var}(B_{N+1}) = p_1 c N^3 R_s^3 + p_2 N R_s^{-1} \qquad (8)$$

and if half practical back rotation is used for denoising b-coefficient, which reduces the transfer noise by a factor of 4 [6] and does not affect the direct noise, the residual noise variance will be

$$\mathrm{Var}(B_{HBR}) = p_1 c N^3 R_s^3/4 + p_2 N R_s^{-1} \qquad (9)$$

Eq. 9 and 10 indicates that the overall noise on $B$ is dominated by the direct noise and transfer noise in the low and high symbol rate region, respectively. Once $p_1$ and $p_2$ is obtained, the optimal symbol rate (in terms of minimizing signal noise) is obtained by letting $\frac{d}{dR_s}\mathrm{Var}(B_{HBR}) = 0$.

$$R_s^{\mathrm{opt}} = (4p_2/3p_1c)^{1/4}N^{-1/2} \qquad (10)$$

## Simulation verification

We conducted simulations on a 40 × 50 km EDFA system (fiber loss: 0.2dB/km, dispersion: 4 ps/km/nm). Ideal transmitter, EDFA (Noise Figure, NF = 5dB) and receiver were used in the simulation to focus on ASE noise limitation on the system performance. 2048 path-averaged 1st-order soliton pulses were used to estimate the variance of $B$ during forward propagation and back rotation, where $b$ are independently modulated with a 4A4PSK constellation. The duty cycle was 1/8 to avoid interaction between adjacent pulses. The symbol rate ranges from 0.5 to 3.5 GBaud, leading to the dispersion length ranging from 3940.8 to 80.6 km. The symbol rate range is selected to ensure both stable existence of path-averaged soliton and enough SNR for small noise

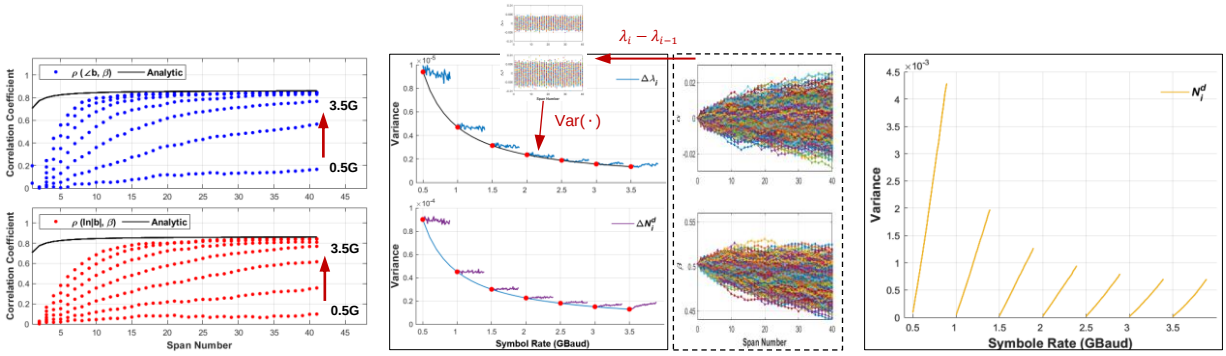


**Fig. 3 (a)** Correlation coefficients between eigenvalue and b-coefficient with different $\boldsymbol{R_s}$; **(b)** Reciprocal dependence of $\boldsymbol{Var(\Delta\lambda_i)}$ and $\mathbf{Var}(\Delta \boldsymbol{N_i^d})$ on $\boldsymbol{R_s}$, **(c)** Illustration of the random walk of $\boldsymbol{\alpha_i}$ and $\boldsymbol{\beta_i}$; **(d)** Linear dependence of $\mathbf{Var}(\boldsymbol{N_i^d})$ on $\boldsymbol{N}$

assumption. NFT was conducted to the coherently received signal after power normalization, and we back rotated with back rotation ratio from 0 to 1 with either ideal or practical back rotation.

Fig. 3(a) shows the correlation coefficient between $\alpha_i$, $\ln|b_i|$, and $\beta_i$, $\angle b_i$, during forward propagation, and the analytic correlation coefficient [6] (ignoring the direct noise) is given for reference. The dominance of the transfer noise gradually increases during transmission, which is quite fast with high symbol rate due to correspondingly low dispersion length. For the 0.5 GBaud signal, the noise on $b$ is basically uncorrelated to the noise on $\lambda$ even at the end of transmission (hence back rotation will not have much improvement on it). In contrast, the correlation for the 3.5 GBaud signal quickly approaches to the theoretical upper limit of 0.87 as the transfer noise quickly dominates.

Fig. 3(b) shows $\mathrm{Var}(\Delta\lambda_i)$ and $\mathrm{Var}(\Delta N_i^d)$ during propagation for different symbol rates. Here each short curve is equivalently an inset showing the noise variance vs span number, similarly, the curves in Fig. 4 are inset of back rotation for each symbol rate. The inset example shows $\Delta\alpha_i$ and $\Delta\beta_i$ during propagation in the 2 GBaud case, which is obtained by differentiating their evolution traces in Fig. 3(c), and then translated into $\mathrm{Var}(\Delta\lambda_i)$. $\mathrm{Var}(\Delta\lambda_i)$ and $\mathrm{Var}(\Delta N_i^d)$ stays basically on the same level during transmission, resulting from the i.i.d ASE loading at each EDFA. The reciprocal dependence of them on $R_s$ is clearly indicated by the fitting curve. From the fitting curve, we obtain $p_1 = 4.8\mathrm{E}3$, $p_2 = 4.5\mathrm{E}4$. In Fig. 3(d), the simulation results verified the assumption that $\Delta N_i^d$ are uncorrelated so $\mathrm{Var}(N_i^d)$ grows linearly with span number, the gradient is nothing but $\mathrm{Var}(\Delta N_i^d)$ as described in Eq. 7.

With $p_1$, $p_2$ obtained, we then verify Eq. 8 and 9. We plot the variance curve of $B_k^{BRI}$ and $B_k^{BRP}$ for each $R_S$ in Fig. 4. Thus, the end point of $B_k^{BRI}$ is $\mathrm{Var}(N_i^d)$, the start and middle point of $B_k^{BRP}$ are $\mathrm{Var}(B_{N+1})$ and $\mathrm{Var}(B_{HBR})$, respectively. They are well matched with the values correspondingly predicted by Eq. 7-9, as shown in Fig. 4. and the optimal symbol rate $R_s^{\mathrm{opt}}$ is evaluated to be 1.25 GBaud. Such a low value may reveal why the symbol rate of discrete NFDM system has long been low. Note that the NF of EDFA, constellation size is not shown in the analysis, so the result should be independent of them. To verify this, we conduct another simulation with 4A8PSK and NF = 3dB, and we found the ration of $p_2/p_1$ is almost the same, and hence the same optimal symbol rate, which implies a deeper relationship between the direct noise on $\lambda$ and $B$.

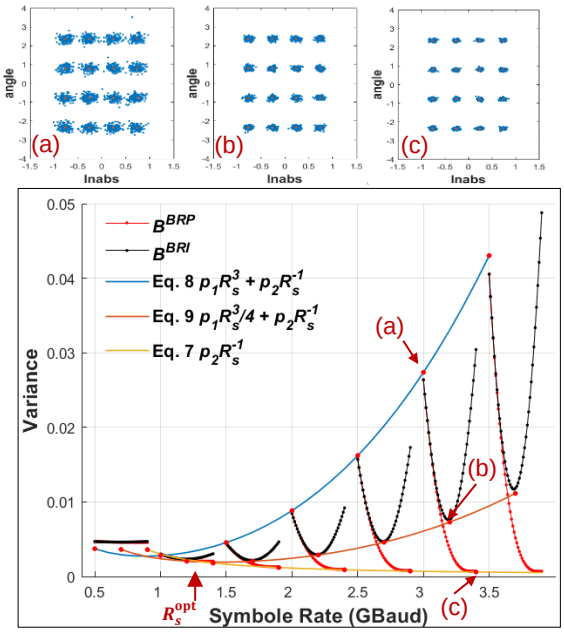


**Fig.4.** The noise variance of $\boldsymbol{B}$ with zero backrotation, half practical back rotation and full ideal back rotation is well fitted by Eq. 8, 7 and 9. Insets: signal distribution of 2GBaud.

The yellow line (Eq. 7) may be interpreted as the best performance, if possible, for a discrete NFDM system, and the orange line (Eq. 9) represents the performance under linear equalization (e.g. half back rotation or LMMSE). Thus, the space in between characterizes the possible improvement of advanced equalization schemes such as nonlinear filter or machine learning.

## Conclusions

We analyze the noise accumulation for a discrete NFDM system and show its dependence on symbol rate. The cubic dependence on symbol rate and transmission distance puts a strong limit on the performance of discrete NFDM system. The optimal symbol rate is found to be low.

## Acknowledgements

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